

Harshita Koppaka

Frisco, TX | harshita.koppaka@gmail.com | linkedin.com/in/harshita-koppaka | github.com/harshitakoppaka

SUMMARY

Software engineering student with applied AI experience building and operating scalable backend and full-stack systems using Python and Java. Strong foundation in data structures, algorithms, and distributed systems, with hands-on experience deploying, debugging, and optimizing AI-powered services. Experienced with cloud platforms, CI/CD pipelines, and reliability-focused engineering in production environments.

EDUCATION

University of North Texas

Bachelor of Science in Computer Science

Denton, TX

Expected May 2026

EXPERIENCE

Bioinformatics Research Assistant

Aug. 2025 – Present

University of North Texas

Denton, TX

- Developed and maintained Python-based backend pipelines for ingesting, processing, and validating multi-terabyte datasets in distributed environments
- Optimized critical code paths using efficient data structures and algorithms, reducing processing time by 40%
- Built, debugged, and supported production-grade services, resolving live system issues and improving reliability
- Designed modular, reusable components using object-oriented principles and version-controlled development practices
- Collaborated with cross-functional researchers and engineers to translate requirements into scalable software solutions

PROJECTS

AI-Enabled Retrieval-Augmented Generation System | Python, FAISS, Transformers

- Built and deployed an AI-powered backend system enabling semantic search over structured datasets using a RAG architecture
- Designed scalable ingestion, preprocessing, and embedding pipelines for production-style AI workloads
- Evaluated retrieval quality and optimized embedding performance for low-latency inference
- Focused on reliability, monitoring, and performance optimization of AI services

Flight Delay Analytics Platform | Python, SQL, Docker, CI/CD

- Engineered distributed backend services processing 50M+ records using scalable Python and SQL pipelines
- Implemented automated testing and CI/CD pipelines to ensure deployment quality and operational reliability
- Troubleshoot data and system issues to improve robustness of production analytics workflows

AI Query Management Service | Java, Python, AWS, REST APIs, HTML, CSS

- Developed and deployed Spring Boot-based backend services supporting AI-enabled application workflows
- Designed validated RESTful APIs with structured responses, logging, and operational visibility
- Deployed services on AWS and integrated automation to improve deployment, monitoring, and developer productivity

TECHNICAL SKILLS

Languages: Python, Java, SQL, C++

AI & Applied ML: Retrieval-Augmented Generation (RAG), Transformers, Embeddings, Model Evaluation

Web & Backend: REST APIs, Distributed Systems Concepts, Microservices, Spring Boot, React, HTML, CSS, JavaScript

Databases: MySQL, MongoDB, DynamoDB, Snowflake

Cloud & DevOps: AWS (S3, Lambda, EC2, Glue, DynamoDB), Docker, CI/CD, GitHub Actions

MLOps & Reliability: Deployment pipelines, performance monitoring, debugging, automated testing

Developer Tools: Git, GitHub Copilot, Linux

HONORS & ACTIVITIES

AWS Certified Solutions Architect – Associate

Dean's List — Spring 2023

HackUNT 2025 Participant — Built an AI-powered academic assistant

Technical Writer on Medium — medium.com/@harshitakoppaka